

Peter Wang

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EDUCATION

University of California, Berkeley (GPA: 3.6/4.0)

May, 2026

B.S. in Computer Science

- **Honors:** Jim & Donna Gray Endowment Award (\$10,000), Leadership Scholar (\$2,000), IEEE Member
- **Relevant Coursework:** Computer Architecture, Microelectronic Devices and Circuits, Machine Learning, Probability & Random Processes, API Development, Data Structure & Algorithms, Integrated Circuits

SKILLS

- **Languages & Frameworks:** Java, Python, C/C++, RISC-V, CUDA, Go, SQL, Matlab, JavaScript, Node.js, JUnit
- **Libraries:** PyTorch, pandas, NumPy, SciPy, Matplotlib, Seaborn, Scikit-learn, Pydantic, SQLAlchemy
- **Hardware:** ASIC, RTL, Floorplan, Code Coverage, Design Verification, Oscilloscope, Spectrum Analyzer

EXPERIENCE

SoC Architecture Intern | Arm

May, 2025 - Aug. 2025

- Constructed a graph-based dataset from RTL to model inter-IP connectivity, congestion, and protocol types (e.g. AXI, CHI) for physical design optimization.
- Prototyped a reinforcement learning macro-placer in PyTorch to reduce congestion and wirelength in SoC floorplan.

Computer Science Peer Advisor | UC Berkeley - College of Engineering

Aug. 2024 - Dec. 2024

- First official hire as a peer advisor for EECS department at Cal Berkeley (eecs.berkeley.edu/cs/advising/#peer)

AI/ML Intern | Honda Research Institute

Jan. 2024 - May. 2024

- Applied L2 Ridge Regression with hyperparameter tuning to address multicollinearity in telematics data.
- Utilized K-means clustering to analyze driving patterns, reducing urban congestion.
- Conducted [exploratory data analysis](#) with NumPy and SciPy, identifying key energy inefficiencies.

Machine Learning Engineer | IEEE

Aug. 2021 - Jun. 2023

- Published research paper "[Finger-Vein Recognition using a NASNet with a Cutout](#)" in 2021 ISPACS, IEEE.
- Designed Python ML models based on the NAS framework (Gradient-based Optimization).
- Utilized CUDA for parallelized training, implementing custom CUDA kernels to accelerate convolutional layers and matrix multiplications, reducing model training time by 36%.

Software Engineer Intern | Blockchain Accelerator

Jun. 2021 - Aug. 2022

- Developed scalable RESTful APIs with Node.js to enhance data processing workflow.

HARDWARE + NLP

Branch Predictor Design

- Developed a hybrid branch predictor (Alpha 21264-inspired) with 4096-entry global, 1024-entry local, and 4096-entry choice predictors, cutting misprediction rates (e.g., Towers: 0.092 to 0.082).
- Implemented a perceptron predictor using weighted history, reducing misprediction by ~13% (e.g., QSort: 0.176 to 0.153) in synthetic correlated-branch tests.

AWS Trainium Accelerator Kernel Developer

- Programmed feedforward neural network (FFNN) and 2D convolution (Conv2D) kernels on AWS Trainium hardware using the Neuron Kernel Interface (NKI) in Python.
- Applied hardware-specific optimizations e.g. engine-aware computation mapping and memory scheduling.

MOSFET Common-Source Amplifier Design

- Analyzed MOSFET characteristics and biasing to achieve optimal amplification in the saturation region.
- Simulated the circuit using LTSpice to determine voltage gain, frequency response, and bandwidth.

Cache System Analysis

- Measured memory access cycles to deduce cache size, associativity, and replacement policies (LRU, FIFO, etc.).
- Tested prefetching strategies such as adaptive prefetching, optimizing spatial and temporal locality.